PHYSWEEP: Does a Video Generator Realize the Physics You Ask For?

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Abstract

Image-to-video generators are often credited with absorbing physical dynamics as implicit world models, a claim the community currently checks with plausibility scores that ask whether a clip looks consistent with real-world motion. Plausibility is the wrong test on its own, because a clip can look natural while encoding the wrong value of the governing physical parameter, and no existing benchmark measures this gap directly. PHYSWEEP *closes it with a fixed, label-free audit, treating a frozen generator as a black box, recovering the realized parameter from generated pixels, and reporting how often generation is trackable at all, how far the realized value sits from the requested one, and which, if either, of the literature's two proposed failure mechanisms the data support. A deterministic-simulator positive control confirms every score is exactly checkable. Applied to three open generators across six sweep axes,* PHYSWEEP *finds a specific, reproducible, previously undocumented failure. Conditional on producing trackable motion, two of the three generate confident, well-fit dynamics that converge to one of a small number of fixed, wrong values selected by the* sampling seed *rather than by the request, reproducing across two independent model families, two physical systems, and an independent tracker. It matches neither the prior reversion nor the case-based clamping the literature anticipates, because the reversion target is seed-conditional rather than a single global default, and a leave-one-out selection rule rejects both; the in-range faithfulness slope is statistically indistinguishable from zero wherever a response is estimable at all. A benchmark averaging over seeds would never see this: each sample is confidently locked to a wrong constant, exactly the failure a plausibility score is structurally blind to. We release the protocol, suite, and analysis code as a reusable, model-agnostic audit for generators that claim physics controllability.*

1. Introduction

Progress in video generation has revived an old hope: that a model trained only to predict pixels will, as a by-product, learn the laws that govern how the world moves [1, 12]. The community has accordingly invested in measuring the *physical plausibility* of generated video, through benchmarks that ask human raters or VLM auto-raters whether motion looks right [8, 9, 15, 27, 29, 45].

Plausibility is the wrong target on its own. A generator can produce a clip that is visually flawless yet encodes an incorrect value of the relevant parameter: a ball can fall along a smooth arc whose implied gravitational acceleration is simply wrong, and no plausibility rater will object, because the arc is locally consistent. The meaningful question is whether a generator realizes the *specific* physics implied by its conditioning, and whether that realization extends past the values the model saw most often in training. The second clause matters because a model that has internalized a law should treat the controlling parameter as a free variable, whereas one that has memorized typical motions falls back on what it has seen once pushed away from the common case.

Recent roadmaps name this gap directly: surveys identify *intrinsic faithfulness* and *controllability* as the frontier and note that metrics for them remain scarce [18, 43], judge-based scores are hard to audit because a judge's weights and APIs drift while a parameter recovered by a fixed estimator does not (Section 2), and the gap is not incidental, since generators optimized for media prioritize visual smoothness over physical fidelity [3]. What is missing is a fixed, auditable measurement of whether a *conditioned* parameter is realized in the pixels.

We turn "physical understanding" of a pretrained generator into a falsifiable measurement. Treat the frozen model as a black-box map from a conditioned parameter θ to a trajectory, recover the realized $\hat{\theta}$ from generated pixels with no human labels, and quantify faithfulness, out-of-range behavior, and the *form* of any failure. PHYSWEEP also adjudi-

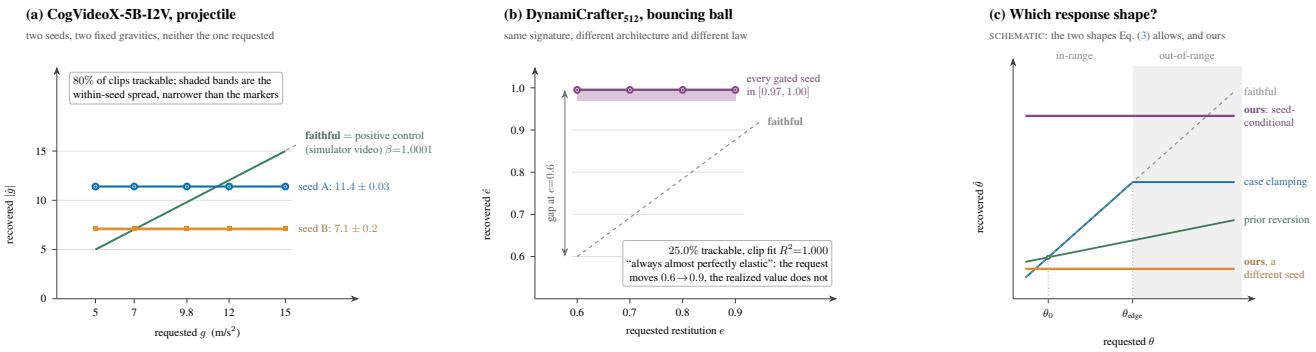


Figure 1. **The seed picks the physics, and it is not tracking noise. (a)** CogVideoX-5B-I2V, projectile, in-range grid, text conditioning: recovered $|\hat{g}|$ is flat in the request and separated by *sampling seed*, with within-seed spread narrower than the plotted markers, while the same tracker and fitter recover the swept parameter from simulator video at slope 1.0001. Each point is a mean over that seed's gated clips at that g (80% trackable in-range), not a single clip. **(b)** The same signature in a different architecture on a different law: DynamiCrafter returns near-perfect elasticity whatever restitution is asked for, at a fit quality that rules out a tracking artifact (25.0% trackable in-range; one fitter-degenerate outlier, $\hat{e} > 1$, physically impossible for a restitution coefficient, is excluded from the plotted line). **(c)** Schematic, drawn to obey Eq. (3): case clamping tracks the identity until θ_{edge} then goes flat, prior reversion is a straight line of slope α crossing the identity at θ_0, both defined in Section 4 as functions of θ alone, so neither can express a response flat in θ and offset by seed, which is why the selection rule rejects both (Proposition 1). Sign convention in (a) is downward-negative; magnitudes are plotted.

cates between the two mechanisms the literature proposes: collapse to a single global default, or case-based mimicry of the nearest seen value. Our delta against the closest prior art (Section 2) is a conjunction: a *swept* control parameter rather than a fixed default, on *frozen black-box* models, over *six axes* rather than gravity alone, with a continuous *parameter-recovery* readout rather than a binary or judge score. We do not claim to be first to recover physics from generated pixels. What is new is the *determinant*: the sampling seed is known to fix a generator's appearance [37], and we show it also fixes the physics, overriding an explicit request for it.

Contributions.

1. **A reusable audit artifact.** PHYSWEEP ships deterministic parameterized systems with ground-truth θ, a fixed label-free estimator, and analysis code (Section 5), validated by a simulator-video positive control. The estimator is fixed rather than learned, so the measurement is auditable by construction (unlike evolving VLM-judge scores [25]), and it applies to any generator accepting one of our two conditioning channels, including ones claiming physics controllability.
2. **Label-free metrics and a mechanism adjudication.** A trackability rate, a Parameter Recovery Error (PRE) with its Extrapolation Gap (EG) and a monotonicity score, two competing failure-form indices, Prior-Reversion (PRI) and Case-Based (CBI), and a leave-one-out rule that lets the data choose between global-default collapse and nearest-value mimicry, or reject both (Section 4) – together turning "did the generator learn physics" into a falsifiable, pre-registered test rather than a qualitative impression.
3. **A formal account of the failure we find, and the statistic that identifies it.** We define seed-conditional defaults and prove they force PRI $\to 1$ with a vanishing held-out R^2 (so rejecting both prior hypotheses is a derived prediction, not an unexplained negative), and that the same signature arises from noise, so PRI alone cannot establish the mechanism, and give a Seed Determination Index that can (Section 4.1). This closes a real gap in how the field reports results: an experiment that is not seed-resolved cannot tell seed-locking apart from a well-behaved generator with noisy output, and we show that distinction is exactly what is at stake here. We report what the index isolates on three frozen generators (Section 6), extending the known imprint of the sampling seed on generated *appearance* [37] to generated *dynamics*.

2. Related Work

A large family of benchmarks scores whether generated video looks physical: human and learned raters of physical commonsense [8, 9], broader law taxonomies [15, 27], real footage [29], conservation-law metrics [45], first-principles organization [16], likelihood on valid-versus-invalid pairs [42], and Physics and Controllability scored by VLM proxies [4, 47]. These ask whether a clip looks right, not whether it realizes a *requested* parameter. Lin et al. [25] argue such judges are hard to reproduce because they evolve, and an evaluation on real systems reports a fine-tuned judge scoring clean recordings only moderately [2]; our plausibility

contrast is an independent instance on synthetic stimuli. Closer to our own readout, Li et al. [24] measure falling dynamics in generated video to drive physics post-training, and Thozhiyoor et al. [32] recover an effective gravity from pixels, show that current generators under-accelerate toward a fixed sub-Earth value, and eliminate metric-scale and frame-rate confounds with a unit-free two-object timing test; Le et al. [23] use a similar readout as a verifiable post-training reward. This is the closest prior art and we do not claim priority over it, but all three fix the target law to gravity and characterize a model's default behavior, whereas our question is orthogonal and complementary: given a parameter the user *asks for*, across six sweep axes, does the realized value move with the request. Their fixed-θ result and our swept-θ result answer different halves of the same worry.

A separate line establishes that the initial noise fixed by the sampling seed leaves a strong, recoverable imprint on what a diffusion model produces [37], but for generated *appearance*; Definition 2 extends this to *dynamics*, and to our knowledge the seed has not previously been shown to fix a quantitative physical parameter. A further wave of 2025–2026 methods claims physics controllability through editable simulation, latent dynamics, geometry guidance, force-vector or explicit-parameter conditioning, and inference-time extrapolation [5, 14, 30, 31, 35, 38, 41, 46]; PHYSWEEP asks what a *frozen* model does beforehand and supplies the fixed measurement against which such claims can be re-checked without training access. Sec. S6 extends this section with the mechanism literature [20, 22], adjacent property-readout and VLM-based work [13, 26, 28, 39, 44], and classic prediction/planning benchmarks [7, 10, 19, 33].

3. Problem: Plausibility versus Parameter Faithfulness

Let a system be governed by a known law f with scalar control parameter $\theta \in \Theta \subset \mathbb{R}$ (the framework extends to vector θ; we sweep scalars for clean identifiability). Given s_0 the true trajectory is $\tau_\theta = f(s_0, \theta)$. A generator G receives conditioning $c(s_0, \theta)$ rendered from τ_θ and produces $v = G(c(s_0, \theta))$; we assume no access to its weights, data, or activations. An external estimator R maps generated pixels to $\hat{\theta} = R(v)$ by tracking the object and fitting f.

Definition 1 (Parameter faithfulness)**.** *G is ϵ-faithful on a sweep $S \subseteq \Theta$ if $\mathbb{E}_{s_0} |R(G(c(s_0, \theta))) - \theta| \leq \epsilon$ for all $\theta \in S$.*

Plausibility, by contrast, is a property of v alone: a rater $P(v)$ scores whether v looks physical on some fixed scale (we use 1–5), independent of θ. The two come apart precisely when v is locally smooth but encodes the wrong θ.

Two hypotheses for out-of-range failure. The literature offers two accounts of what a generator does when θ leaves the common range S_{in}, predicting different $\hat{\theta}$ that a sweep can separate. Under **prior reversion**, $\hat{\theta} \approx \alpha\theta + (1-\alpha)\theta_0$ with a single fixed default θ_0 such as Earth gravity, independent of where in S_{out} the request lies. Under **case clamping** [20], $\hat{\theta}$ sticks at the nearest in-range value seen, θ_{edge}. The two coincide only when $\theta_0 = \theta_{\text{edge}}$. Both assume $\hat{\theta}$ is a function of θ alone; Section 4.1 shows our data violate that assumption and derives the consequence.

4. Metrics

Fix a system with sweep grid $\{\theta_i\}_{i=1}^n$ partitioned into an in-range set S_{in} (values common in natural video) and an out-of-range set S_{out} (values rare or absent). For each θ_i we draw m seeded initial conditions and generations and obtain recovered values $\{\hat{\theta}_{ij}\}_{j=1}^m$.

Parameter Recovery Error and Extrapolation Gap. The normalized recovery error on a set S, with small $\delta > 0$ for stability, and the gap between splits, are

$$\text{PRE}(S) = \frac{1}{|S|} \sum_{\theta_i \in S} \frac{1}{m} \sum_{j=1}^{m} \frac{|\hat{\theta}_{ij} - \theta_i|}{|\theta_i| + \delta}, \tag{1}$$

$$\text{EG} = \text{PRE}(S_{\text{out}}) - \text{PRE}(S_{\text{in}}). \tag{2}$$

Lower PRE is better and $\text{PRE} = 0$ is perfect faithfulness; $\text{EG} \approx 0$ means the model extrapolates as well as it interpolates.

Trackability rate. Not every generation contains motion a law can be fitted to. We report the fraction of clips clearing the fit gate ($R^2 \geq 0.8$) as a metric in its own right, not a silent exclusion: it is label-free and Section 6 shows it carries information an automated plausibility rater does not.

Faithfulness slope and monotonicity. β is the least-squares slope of $\hat{\theta}$ on θ over S_{in}: a faithful model has $\beta \approx 1$, one ignoring the conditioning $\beta \approx 0$. Because the response may be nonlinear we also report Spearman's ρ, capturing whether the ordering of requested values survives when the scale is off.

Two failure-form indices and a selection rule. To adjudicate between the hypotheses of Section 3 we fit both to the out-of-range data:

$$\begin{aligned} \text{prior reversion:} \quad & \hat{\theta} = \alpha\theta + (1-\alpha)\theta_0 + \varepsilon, \\ \text{case clamping:} \quad & \hat{\theta} = \min(\theta, \theta_{\text{edge}}) + \varepsilon, \end{aligned} \tag{3}$$

$$\text{PRI} = 1 - \hat{\alpha}, \qquad \text{CBI} = 1 - \text{RSS}_{\text{clamp}}/\text{RSS}_{\text{null}}.$$

$\text{PRI} = 0$ means the request is followed, $\text{PRI} = 1$ means the output ignores it and returns θ_0, and CBI is the out-of-range

variance explained by clamping at the edge. We select by leave-one-θ-out R^2 and ΔAIC, reporting *neither* when both fail the held-out test. θ_0 and θ_{edge} are reported, not assumed.

Stochasticity, scale, and uncertainty. Generators are sampled with $m \geq 5$ seeds per condition. We separate the bias $\mathbb{E}[\hat{\theta}]-\theta$ from the within-condition dispersion $\mathrm{std}(\hat{\theta})$, so an unbiased-but-noisy model is not confused with a biased-but-consistent one. Because we render the conditioning, spatial and temporal scales are fixed within a clip and all metrics use the ratio $\hat{\theta}/\theta$, invariant to a common rescaling; a scale-invariance ablation checks this. Metric-scale ambiguity is a known confound for physics readouts from generated video [32] and this is our defense against it. All metrics carry 95% bootstrap CIs over (θ_i, seed) pairs.

Well-definedness of $\hat{\theta}$. The metrics mean something only if R recovers a well-defined quantity; Sec. S2 gives the two regularity conditions (trackability, identifiability) under which $\hat{\theta}$ converges in probability to the realized parameter, a claim about the *estimator*, not the generator, and discloses one small bias this exposes for the damped pendulum. When the motion is not well described by f the residual is large, so we gate at $R^2 \geq 0.8$ and treat low-fit generations as *non-physical* rather than forcing a $\hat{\theta}$.

4.1. Seed-conditional defaults

Both hypotheses assume the realized value is a function of θ alone. Our data are not, so we state the general model and derive what it implies for the indices rather than treating the resulting "neither" as an unexplained negative. This is our only novel formal claim.

Definition 2 (Seed-conditional default)**.** *Let the sweep be run as a* crossed *design, every grid value with every seed, so that the seed is independent of θ. G has a* seed-conditional default *on S if there is a non-constant $c : \mathcal{Z} \to \Theta$ with $\hat{\theta}(\theta, z) = c(z) + \varepsilon$, where ε has mean 0 and variance σ_ε^2 and is independent of (θ, z), and $\sigma_c^2 = \mathrm{Var}_z[c(z)] > 0$. Prior reversion with a single global θ_0 is the special case $\sigma_c^2 = 0$, $c \equiv \theta_0$, so the model nests it. The map c is the dynamical analogue of the appearance imprint the seed is known to carry [37].*

Proposition 1 (The indices are misspecified, predictably)**.** *Under Definition 2: (i) the pooled least-squares prior-reversion fit has $\hat{\alpha} \to 0$ in probability, so* PRI $\to 1$*; (ii) its implied default converges in probability to $\mathbb{E}_z[c(z)]$, a population mean that no individual generation realizes; and (iii) every predictor that is a function of θ alone has population $R^2 = 0$, so its leave-one-θ-out R^2 converges in probability to 0 and, for any fixed margin $\tau > 0$, a selection rule accepting only $R^2_{\text{LOO}} > \tau$ returns* neither *with probability tending to one. Statements (i)–(iii) hold verbatim when $\sigma_c^2 = 0$ and σ_ε^2 is large, so* PRI *and the selection outcome cannot distinguish a seed-conditional default from unstructured noise.*

Sketch. The crossed design gives $\hat{\theta} \perp \theta$, hence $\mathrm{Cov}(\theta, \hat{\theta}) = 0$, which yields (i) and, through the intercept, (ii). For (iii) the best θ-only predictor is the marginal mean, so the population R^2 is 0 and the held-out estimate converges to it. The final sentence follows because none of the three arguments uses $\sigma_c^2 > 0$. Full proof in Sec. S3. □

The margin in (iii) is not a detail: with R^2_{LOO} centered on zero its *sign* is noise, so a rule accepting any positive value would admit unstructured data as prior reversion (our rule uses $\tau = 0.1$; justification and a smoke-test check in Sec. S3). The proposition's last sentence is why a further statistic is needed: separating the regimes requires one that conditions on the seed.

Seed Determination Index. Decompose $\mathrm{Var}(\hat{\theta})$ over the design's two factors: FDI is the share from the requested θ, SDI the share from the seed, the latter a one-way random-effects intraclass correlation so it is not upward-biased at this design size. Under Definition 2, FDI $\to 0$ and SDI $\to \sigma_c^2/(\sigma_c^2 + \sigma_\varepsilon^2)$ in probability; when instead $\hat{\theta} = \theta + \varepsilon$ with ε independent of the seed, the two swap. On synthetic ground truth the released implementation returns SDI $= 0.999$ for a seed-locked generator against 0.000 faithful, 0.000 pure noise and 0.060 single global default, and refuses rather than guessing below two seeds.

5. Suite, Pipeline, and Protocol

Each system is a deterministic renderer producing conditioning frames, ground-truth θ, and one interpretable sweep axis. Scenes are simple and high-contrast so the tracker is robust and the identifiability conditions above hold easily, keeping the measurement rather than the perception in focus. That has a cost Section 7 takes seriously: minimalist synthetic scenes sit far from every tested model's training distribution. Five systems supply six axes, the pendulum contributing two.

System	θ	Recovered statistic	Sweep grid: in-range / out-of-range
Projectile	g	parabola curvature	$\{5, 7, 9.8, 12, 15\}$ / $\{1.6, 3, 20, 25\}$
Damped pendulum	ω	zero-crossing period	$\{2, 3, 4, 5\}$ / $\{0.5, 1, 8, 12\}$
	ζ	envelope decay	$\{.02, .05, .1\}$ / $\{.3, .6, .9\}$
Bouncing ball	e	bounce-height ratio	$\{.6, .7, .8, .9\}$ / $\{.2, .35, .97\}$
Spring–mass	k	oscillation period	$\{20, 40, 60, 80\}$ / $\{5, 10, 200, 400\}$
Inclined slide	μ	along-slope acceleration	$\{.1, .2, .3, .4\}$ / $\{.01, .7, 1.0\}$

Table 1. **The PHYSWEEP suite.** Five deterministic renderers give six sweep axes, each with a closed-form map from a tracked statistic to the parameter, so recovery is exact and label-free. Units: g m/s^2, ω rad/s, k N/m; e, ζ, μ dimensionless. Rendering settings and per-pair seed counts are in Sec. S4.

Pipeline. Figure 2 sketches the loop described here. For each grid value θ_i and seed z_j we render conditioning

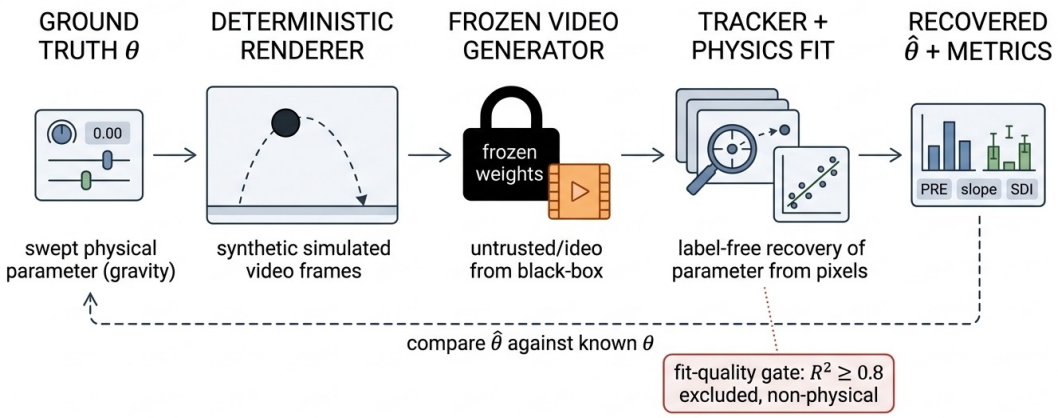


Figure 2. **The measurement pipeline.** A known θ drives a deterministic renderer, whose conditioning frames are handed to the frozen generator under audit; the resulting clip is tracked and fit in closed form to recover $\hat{\theta}$, which is compared back against the known θ to compute every metric in Section 4. The fit-quality gate ($R^2 \geq 0.8$) is the only point where a generation can be excluded, and it is excluded as *non-physical*, not silently dropped. Only the generator (dark box) is untrusted; every other stage is exact and label-free by construction.

frames, generate a continuation, track the observable and fit f in closed form to obtain $(\hat{\theta}_{ij}, R^2_{ij})$; clips with $R^2_{ij} < 0.8$ are marked non-physical and excluded from PRE, and that rate is reported alongside every result. A VLM rater scores the same clips. Rendering uses a small deterministic 2D engine with no external assets; tracking is color-blob centroiding, with CoTracker3 [21] as an independent second tracker in ablation. Pseudocode is in Sec. S1. The pipeline is inference-only, run on a shared NVIDIA A100-SXM4 (40 GB), not a single consumer GPU (CogVideoX's 25.3 GB peak, with tiling, exceeds a 24 GB budget), for an estimated ~ 57 total GPU-hours across all runs; the estimate's derivation and per-model figures are in Sec. S4.

How θ is conditioned. Models accept different inputs, so we treat the channel as a controlled axis. **Frame-implied conditioning** supplies rendered frames whose motion already implies θ and asks the model to continue, isolating dynamics from semantics. **Text-specified conditioning** supplies a first frame plus a prompt naming the regime, for example on the Moon versus on Jupiter, testing whether a named control moves the realized parameter. Section 6 reports which channel each result uses, since only one model admits both.

Protocol. We tested three frozen open generators, each within reach of a single high-end GPU (VRAM figures in Sec. S4): LTX-Video [17], CogVideoX-5B-I2V [40], DynamiCrafter$_{512}$ [36]. Stable Video Diffusion [11] was dropped before any run because its pipeline accepts no text prompt; a quantized Wan [34] was considered and not run. Plausibility is contrasted against, not used as, the main metric via one VLM auto-rater [6]; *no human subset and no* VBENCH-2.0 *score were collected.* Before running we pre-committed which of four outcomes would confirm or refute each hypothesis, whatever the sweep actually showed: (1) faithful in-range, fails out-of-range, prior reversion or case clamping selected – the field's own anticipated headline; (2) no conditioning effect ($\beta \approx 0$); (3) faithful even out-of-range (EG $\approx$ 0), a surprising positive result reported as such; or (4) neither hypothesis fits. Whichever occurred would drive the abstract, title, and contributions; Section 6 names which two of the four the sweep produced. We use $m \geq 5$ seeds per (θ_i, model), report 95% bootstrap CIs and effect sizes rather than p-values alone, test $\text{PRE}(S_{\text{out}}) > \text{PRE}(S_{\text{in}})$ by paired bootstrap, and report held-out R^2 and ΔAIC for selection. Model repositories and licenses are in Sec. S4, which also discloses that no explicit commit/revision was pinned when loading any model, so each resolved to that repository's default-branch head at run time; Sec. S5 is the reproducibility checklist.

6. Results

Two protocol deviations are stated once and carried through. *First*, every result except the conditioning-channel ablation uses **text-specified conditioning**: LTXConditionPipeline, the `diffusers` component giving first-and-last-frame conditioning for LTX-Video, never computes the scheduler's required shift parameter in the release used here and, once patched, still produces incoherent output regardless of resolution or scene content, confirmed against the working single-image pipeline on an identical input; CogVideoX's official pipeline accepts only one conditioning image. *Second*, DynamiCrafter$_{512}$ replaced Stable Video Diffusion as above.

Trackability (%)	proj.	slide	ball
LTX-Video	**20**	**35**	5
CogVideoX	**80**	0	5
DynamiCrafter	**64**	**80**	25

Table 2. **Coverage on the three axes short clips can identify at all**: fraction of in-range clips whose tracked trajectory clears the $R^2 \geq 0.8$ fit gate, under text conditioning. Shading is proportional to trackability, gray marks a measured zero (CogVideoX, inclined slide), and **bold** marks the five cells yielding enough gated clips for a slope. The remaining three axes are frame-budget-bound rather than a conditioning result (Section 7) and are omitted here.

The instrument works: simulator-video positive control. Ground-truth simulator continuations, not model output, through the identical tracker and fitter, pooled across all six axes and both splits (203 measurements), give $\mathrm{PRE}(S_{\text{in}}) = 0.0031$ (CI $[0.0019, 0.0045]$), slope $\beta = 1.0001$ (CI $[1.00008, 1.00023]$), and mean fit $R^2 = 0.986$. The estimator recovers a swept parameter when one is present, so a null slope on model output is a property of the model or the stimulus, not the pipeline.

Conditioning does not move the realized parameter. Table 2 shows the three axes short clips can identify at all; trackability never exceeds 80% there. The nine cells on the other three, frame-budget-bound axes (not shown in the table) are a separate, architectural 0% (Section 7), so ten of all eighteen model-axis pairs measure 0% trackability in total, for two different reasons. Only five cells support a slope estimate, and every one is indistinguishable from zero while excluding $\beta = 1$: LTX-Video projectile -0.015 $[-0.090, 0.039]$ and inclined slide -0.464 $[-1.751, 1.097]^{\dagger}$; CogVideoX projectile 0.070 $[-0.168, 0.305]$; DynamiCrafter projectile -0.025 $[-0.466, 0.480]$ and inclined slide 0.521 $[-6.185, 6.850]^{\dagger}$. The two marked $\dagger$ also contain $\beta = 1$, showing lack of power rather than absence, so we do not count them as evidence for the null; the remaining three are inconsistent with faithful conditioning. Recovery error is correspondingly large on the projectile cells ($\mathrm{PRE}(S_{\text{in}}) = 0.853, 2.016, 0.781$ for LTX-Video, CogVideoX, DynamiCrafter, against 0.0031 for the positive control), and rank order fares no better: Spearman's ρ over the full projectile sweep is -0.268 (LTX-Video) and 0.046 (CogVideoX), so not even the *ordering* of requested values survives.

The failure is seed-locked, not untrackable noise. The three models reach the same null slope through visibly different behavior, confirmed by frame inspection. **LTX-Video** mostly fails to generate any real fall: the conditioned disk stays near its initial position regardless of the requested g or how the prompt describes it, tested up to "extremely strong gravity, like Jupiter, violent rapid acceleration downward." **CogVideoX and DynamiCrafter** instead generate confident, well-tracked motion that is simply the wrong amount, with recovered $\hat{g}$ clustering tightly by sampling seed and close to independent of the request: on CogVideoX one seed gives $\hat{g} \approx -11.4 \pm 0.03$ across every in-range $g \in \{5, 7, 9.8, 12, 15\}$ and another gives -7.1 ± 0.2 (Figure 1), reproducing out-of-range on the same seeds.

That second mode also rules out the reading that the trackability rate invites, that these models simply cannot render minimalist synthetic scenes and the survivors are flukes: if the stimulus were merely un-renderable, the recovered values clearing the gate would be scattered, not tight. They are not (within-seed dispersion 0.03–0.13, individual-clip fit R^2 up to 1.000, clusters reappearing under an independent tracker). Confident, well-fit, seed-locked motion is what a model ignoring θ produces, not what one failing to render produces, so we state the headline conditionally: **conditional on producing physically trackable motion, none of the three models detectably honors a text-specified physical parameter.** Trackability bounds the *coverage* of that claim, not its validity.

The signature crosses architectures and physical systems. The pattern is not specific to the projectile. LTX-Video's near-total-inertia failure generalizes, with four of its five remaining axes below 6% trackable. CogVideoX's seed-determined convergence generalizes only where clips survive the gate, which fails entirely on spring-mass and inclined slide; on bouncing-ball, every gated seed converges to a recovered restitution in $[0.97, 1.00]$ regardless of the request, and DynamiCrafter independently reproduces the identical signature on the same system ($\hat{e} = 0.99$–1.00 across the full grid, individual-clip $R^2 = 1.000$): two architectures landing on the same wrong-but-confident behavior. Three of DynamiCrafter's axes (ω, ζ, spring-mass) are separately bounded by its fixed 16-frame output, too short for one oscillation period, a capability ceiling we keep distinct from a conditioning failure throughout.

A third variant: a split, not a single, default. DynamiCrafter's inclined slide adds a variant the two-hypothesis framework does not name. One seed behaves like the single-default cases above, at $\hat{\mu} \approx 1.80$ across the whole grid; a second shows *two* tight clusters, $\hat{\mu} \approx 4.7$ ($\mathrm{std} = 0.04$) in-range and ≈ 7.6 ($\mathrm{std} = 0.13$) out-of-range. That is neither ignoring θ nor a single global default. We name it the *split-default* pattern.

Neither existing hypothesis explains it, exactly as predicted. We adjudicated every model-axis pair with at least

four gated out-of-range points, yielding **8 of 18** pairs (Table 3), all labeled *neither*: AIC prefers the prior-reversion shape over clamping throughout and PRI is high, yet held-out R^2 is negative for both hypotheses on every pair. By Proposition 1 that is exactly what a seed-conditional default predicts. SDI (Table 3) establishes that reading rather than inferring it, decisively supporting seed-locking on five of the eight pairs but not the remaining three, so we report the signature as established only where SDI says so. The paired-bootstrap test is computable for six pairs and exactly two intervals exclude zero, in *opposite* directions (Sec. S4): CogVideoX/projectile shows the predicted worse-out-of-range pattern, while DynamiCrafter/inclined-slide shows a *lower* out-of-range PRE, a consequence of the split-default pattern rather than faithfulness. Holm-corrected across all six, only CogVideoX/projectile survives at $\alpha = 0.05$ (adjusted $p = 0.033$; every other pair's adjusted $p \geq 0.53$). Ten pairs stay unadjudicated: three hit DynamiCrafter's 16-frame ceiling, the other seven a near-zero gated yield. Of the four pre-committed outcomes (Section 5), the sweep produced a mix of the second (*no conditioning effect*, on the projectile and inclined-slide cells whose slope CIs exclude $\beta = 1$) and the fourth (*neither hypothesis fits*, on all eight adjudicated pairs above), not the field's anticipated headline story or a surprising positive result.

			Extrapolation	Mechanism		
Model	**Axis**	n_{out}	**EG [95% CI]**	**PRI**	**ΔAIC**	**SDI**
LTX-Video	projectile	7	0.985 [−0.33, 2.73]	1.000	−4.0	0.995
LTX-Video	bouncing ball	4	—	1.000	−9.8	0.985
LTX-Video	inclined slide	4	0.264 [−0.51, 1.00]	0.514	−11.3	0.511
CogVideoX	projectile	12	1.068 [0.27, 1.89]*	0.974	−47.9	0.919
CogVideoX	bouncing ball	9	—	0.999	−100.4	0.997
DynamiCrafter	projectile	17	0.050 [−0.22, 0.34]	0.988	−22.2	0.343
DynamiCrafter	bouncing ball	6	−0.761 [−2.74, 0.34]	0.223	−0.6	0.997
DynamiCrafter	inclined slide	20	−0.939 [−1.57, −0.35]‡	1.000	−16.0	0.176

Table 3. **Mechanism adjudication**, every model-axis pair with ≥ 4 gated out-of-range points; all eight select as *neither* prior reversion nor case clamping, as Proposition 1 predicts. EG $=$ $\text{PRE}(S_{\text{out}}) - \text{PRE}(S_{\text{in}})$, undefined (—) on the bouncing-ball rows since θ enters PRE's denominator and the in-range grid never clears the gate at a stable point. PRI $= 1$ means the output ignores the request; $\Delta\text{AIC} < 0$ favours prior reversion over clamping; both use out-of-range data only. *Excludes zero in the hypothesized direction ($p = 0.0055$); ‡excludes zero in the *opposite* direction, which is not faithfulness. SDI is the column that decides the mechanism (read in text above).

A plausibility rater sees none of this. Qwen2.5-VL-7B-Instruct [6] rated all 630 saved generality clips 1–5. Pooled mean P is low for every model (LTX-Video 2.20, DynamiCrafter 1.76, CogVideoX 1.26), which looks consistent with the hypothesis on its face, but it never assigned a 4 and CogVideoX's scores are bimodal (1 or 5, never 2–4). Checked against what it should track, the correlation between tracker fit R^2 and P is 0.075, 0.019 and −0.164 respectively, indistinguishable from zero: a CogVideoX clip whose trajectory collapses into texture noise ($R^2 = 0.48$) scored $5/5$ while a cleanly tracked LTX-Video fall ($R^2 = 0.99$) scored $1/5$ (truncation, parsing failure and file corruption ruled out). This rater does not track physical coherence here, which is why we report trackability as a separate metric.

The result survives every nuisance parameter we varied. **(a) Tracker swap.** CoTracker3 [21] rerun on every clip agrees strongly where both trackers clear the gate (LTX-Video projectile $r = 0.984$, inclined slide $r = 0.997$); DynamiCrafter agreement is strong on the reported seed cluster but disagrees in sign on a second seed, so the phenomenon replicates under an independent tracker while one cluster's value is tracker-sensitive. **(b) Nuisance parameters.** Across five disk-size and frame-rate rescalings, four guidance values, two prompt templates and two conditioning resolutions, the in-range slope stays small and unstable in sign (LTX-Video −0.26 to +1.38 over 13 variants; DynamiCrafter −0.25 to +0.12 over 7, since neither guidance nor resolution applies to it); CogVideoX's columns are uninformative rather than confirmatory, since no clip clears the gate at this sweep's shorter frame budget. **(c) Gate sensitivity.** The null does not flip at any threshold from 0.5 to 0.95.

Changing the conditioning channel does not rescue it. We ran frame-implied conditioning on DynamiCrafter, the one model with a reachable official multi-frame checkpoint, on the three axes where its text-channel data is usable (projectile conditioned on first and *middle* frame, since its true last frame is degenerate by construction and the middle frame's height is confirmed to vary with g). It does not recover faithfulness: every gated projectile clip converges to $\hat{g} \approx 0$ (slope 0.017, CI $[-0.008, 0.047]$), so where the text channel gave seed-diverse defaults, the frame channel collapses to one near-universal default of no fall, not a degenerate-conditioning artifact. Inclined slide is starker still, $\hat{\mu}$ near 1.7–1.8 while true μ runs from 0.01 to 1.00. Dropped rates (pooled, from a separate, larger-seed-count run than Table 2) are lower under the frame channel on all three axes (10% vs. 30% projectile, 23% vs. 29% inclined slide, 63% vs. 86% bouncing ball), so the two channels are not directly comparable cell-for-cell against Table 2, but neither one is faithful.

7. Limitations

Trackability bounds coverage. Trackability never exceeds 80% and is 0% on ten of eighteen pairs. Section 6 argues from the survivors' seed-locked tightness that this is not

simply un-renderable output, which is why the headline is conditional, but what these models do where nothing clears the gate remains open: a contemporaneous study recovering gravity from current generators on photorealistic stimuli obtained usable trajectories where we largely do not [32].

Model currency. The gap is specific rather than generic: we test LTX-Video and CogVideoX-5B while later releases in both families exist, and omit the open models most often credited with physically coherent motion. It also looks closable, since Wan reports an 8.19 GB text-to-video variant at 1.3B [34] plausibly inside our budget, though not the image-to-video checkpoint we would need.

Design gaps, closed and remaining. The pipeline has a *positive* control (simulator-video recovery at $\beta = 1.0001$) and now a *negative* one: permuting θ labels within each model-axis pair and refitting β, 10,000 times on the five slope-estimable pairs, places every observed β well inside its own null distribution (two-sided permutation p from 0.56 to 0.91), calibrating "indistinguishable from zero" against an actual null. A TOST against a pre-specified smallest effect $\epsilon = 0.2$ at $\alpha = 0.05$ (SE backed out from each pair's bootstrap CI) shows only LTX-Video/projectile is formally equivalent to zero ($p_{\text{TOST}} < 0.001$); the other four, including the two inclined-slide pairs whose CIs also contain $\beta = 1$, do not clear the bar (p_{TOST} from 0.14 to 0.64), consistent with underpowered estimates rather than a plausible large effect. So "no detectable response" is absence of evidence, not evidence of absence, at four of these five pairs; only LTX-Video/projectile is positively established. Native clip length is still confounded with model identity (24, 49 trimmed to 24, 16 frames), so cross-model trackability comparisons are not clean.

Channel, coverage, ceilings. All results but the channel ablation use text conditioning; frame-implied conditioning is untested for LTX-Video and CogVideoX for the pipeline reasons above. Adjudication covers 8 of 18 pairs, DynamiCrafter's 16-frame output makes three axes untestable for reasons unrelated to conditioning, and a joint (ω, ζ) pendulum sweep returned 0% trackability (best $R^2 = 0.14$), which we report as uninformative rather than negative.

Plausibility contrast and realism. One rater on one prompt does not track physical coherence here, and we tested no other prompt, larger VLM or human raters. The suite also trades ecological validity for identifiability, so conclusions concern controllable low-dimensional dynamics.

8. Conclusion

We reframe learning physics from video, for a generator, as a falsifiable label-free question of parameter faithfulness, and supply a black-box diagnostic, three metrics with a stated identifiability condition, a controllable suite, and a protocol separating plausibility from correctness. Applied to three frozen open generators it finds that, conditional on producing trackable motion, none detectably honors a text-specified parameter, and that the failure is not merely absent signal: two of three converge to a few fixed, wrong values selected by the sampling seed rather than the request, reproducibly across model families, axes and trackers. The seed is already known to fix a generator's appearance [37]; here it fixes the physics. A benchmark averaging over seeds reports a smooth mean and hides that each sample is locked to a wrong constant, so seed-resolved reporting is not optional. The next targets are whether the signature survives on photorealistic stimuli and on current generators.

Reproducibility, ethics, and use of LLMs. All systems are procedurally generated by a seeded deterministic engine, models are public, trackers and fitting are off-the-shelf or closed-form, and no labels are used; the analysis code and system generators are released as anonymized supplementary material. No human-subjects data was collected and the diagnostic is meant to prevent overclaiming about video generators as physical simulators, which has safety relevance in robotics and science. Per WACV's LLM policy: an assistant drafted the initial consistency proof (Sec. S2), the physics-recovery fitters and prose throughout. Both were verified by the authors before use, the proof by hand against its three stated assumptions (trackability, identifiability, regularity; Sec. S2 of the supplementary material), which surfaced the ω-versus-ω_d correction and one latent code bug, and every fitter against closed-form trajectories with a passing smoke test. A careful line-by-line check of both supplementary proofs (Proposition 1 and the appendix's consistency and misspecification proofs, Sec. S2–S3) was performed during finalization, independently re-deriving the crossed-design independence argument, the OLS-slope and population-R^2 convergence steps, the SDI formula against the standard ICC(1) estimator, and the Definition 2 nesting claim, with no error found. The assistant did not choose hypotheses, design grids or interpret results, and the authors take responsibility for all content.

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